

Task-5-Statistical Analysis Report

Objective:

To analyze the Marketing A/B Testing dataset using statistical methods and determine whether there is a significant difference between the advertisement groups.

Steps Performed:

1. Loaded the Marketing A/B Testing dataset using Pandas.
2. Calculated Mean, Median, and Mode for the Total Ads column.
3. Visualized the distribution using a histogram.
4. Separated the data into advertisement groups.
5. Performed an Independent T-Test on the groups.
6. Calculated the P-value and interpreted the results.

Descriptive Statistics:

Mean: 24.82

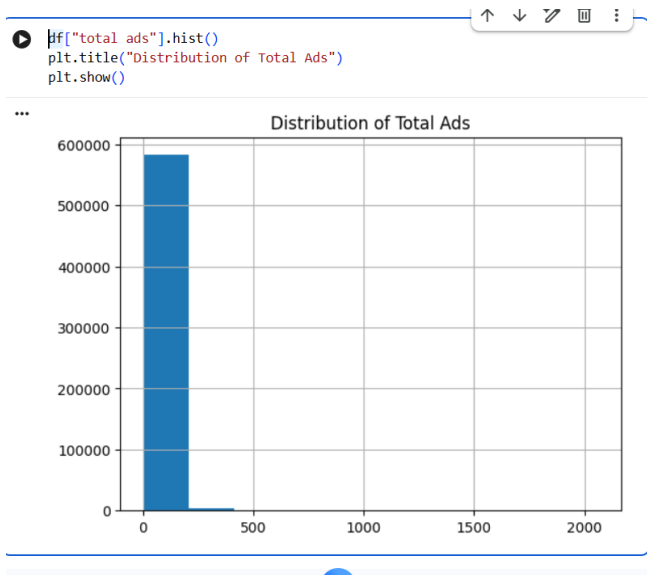
Median: 13

Mode: 1

```
6] mean_ads = df['total ads'].mean()  
0s median_ads = df['total ads'].median()  
mode_ads = df['total ads'].mode()[0]
```

```
7] print("Mean:", mean_ads)  
0s print("Median:", median_ads)  
print("Mode:", mode_ads)
```

```
✓ ... Mean: 24.82087600599217  
Median: 13.0  
Mode: 1
```



T-test

```
group1 = df[df["test_group"]=="ad"]["total ads"]  
group2 = df[df["test_group"]!="ad"]["total ads"]  
t_stat, p_value = ttest_ind(group1, group2)  
print("T-statistic:", t_stat)  
print("P-value:", p_value)
```

T-statistic: 0.21391499762385457
P-value: 0.8306134186810267

P-value

```
if p_value < 0.05:  
    print("Reject Null Hypothesis")  
    print("Groups are significantly different")  
else:  
    print("Fail to Reject Null Hypothesis")  
    print("No significant difference between groups")
```

... Fail to Reject Null Hypothesis
No significant difference between groups

T-Test Results:

T-Statistic: 0.214

P-Value: 0.831

Conclusion:

Since the P-value is greater than 0.05, there is no statistically significant difference between the advertisement groups.